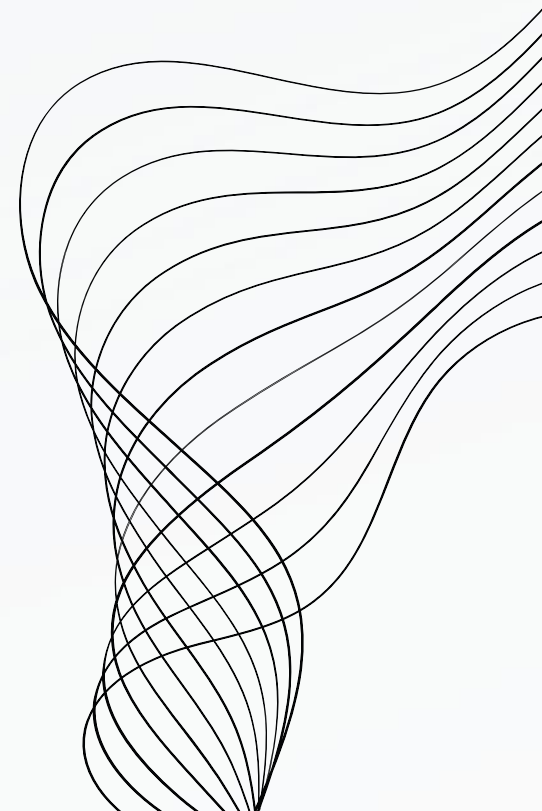


DATAVIS

LECTURE №4. DATA TYPES AND PROPERTIES

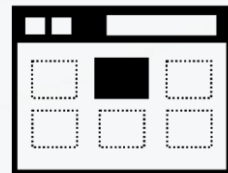


WHAT IS DATA?

Data is the raw information we collect, store, and analyze. Information needed to reach a conclusion or make a decision (Oxford Dictionary).



Numerical



Categorical



Temporary



Text



Geographical

WHY KNOW ABOUT DATA TYPES?

01

THE RIGHT CHOICE OF VISUALIZATION
IMPROVES UNDERSTANDING OF DATA;

02

HELPS TO AVOID DISTORTION OF
INFORMATION;

03

ALLOWS FOR MORE EFFECTIVE
COMMUNICATION OF ANALYSIS RESULTS;



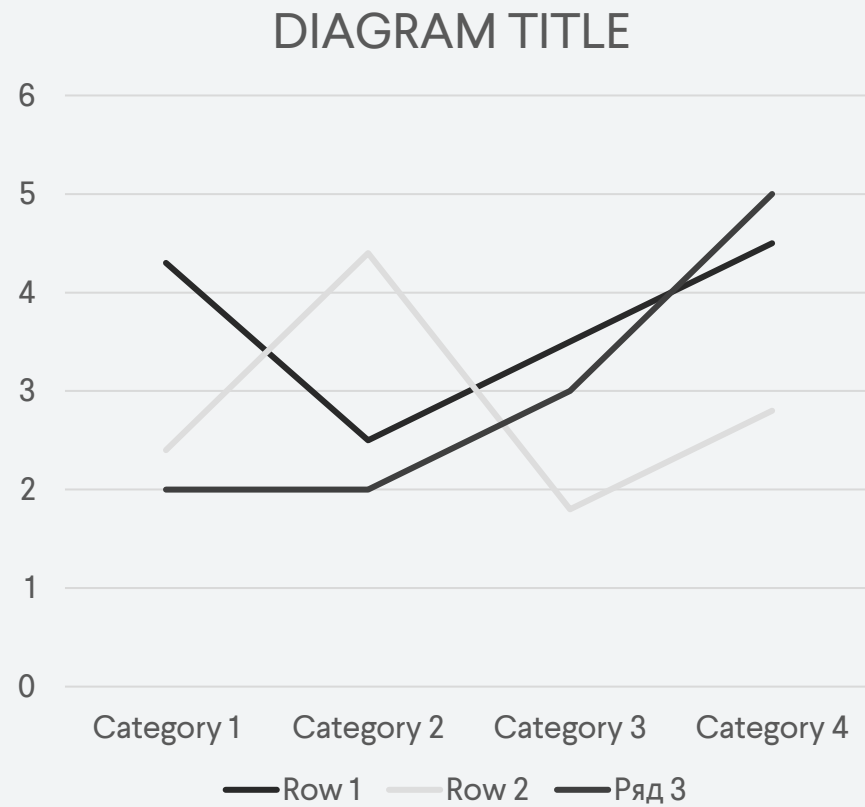
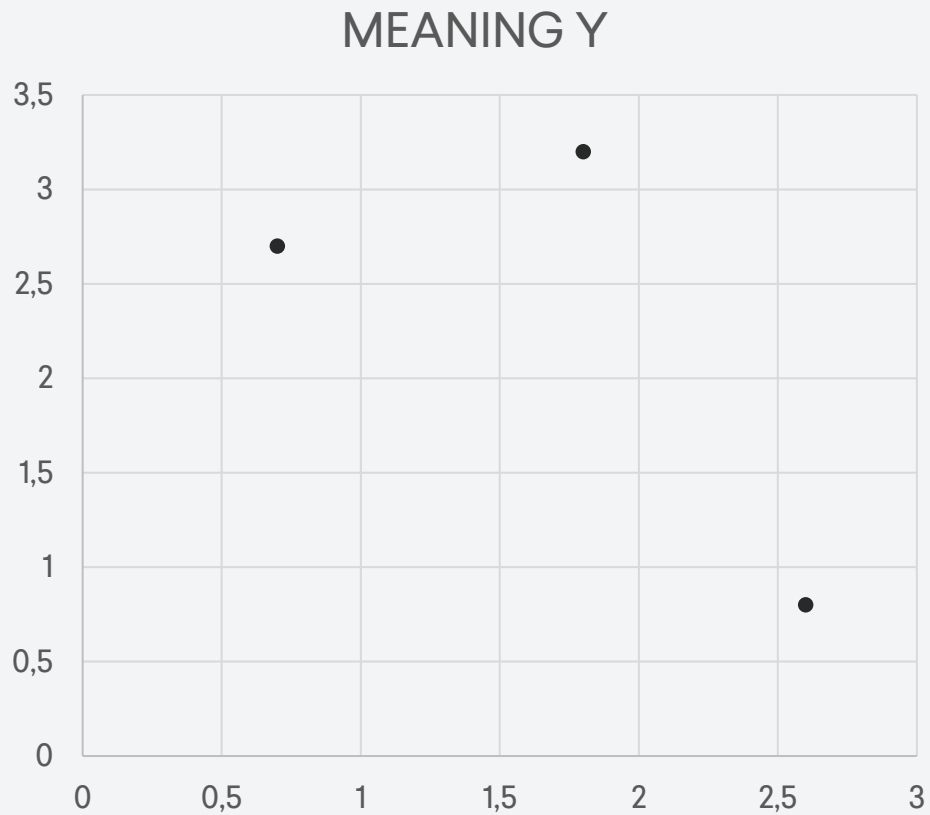
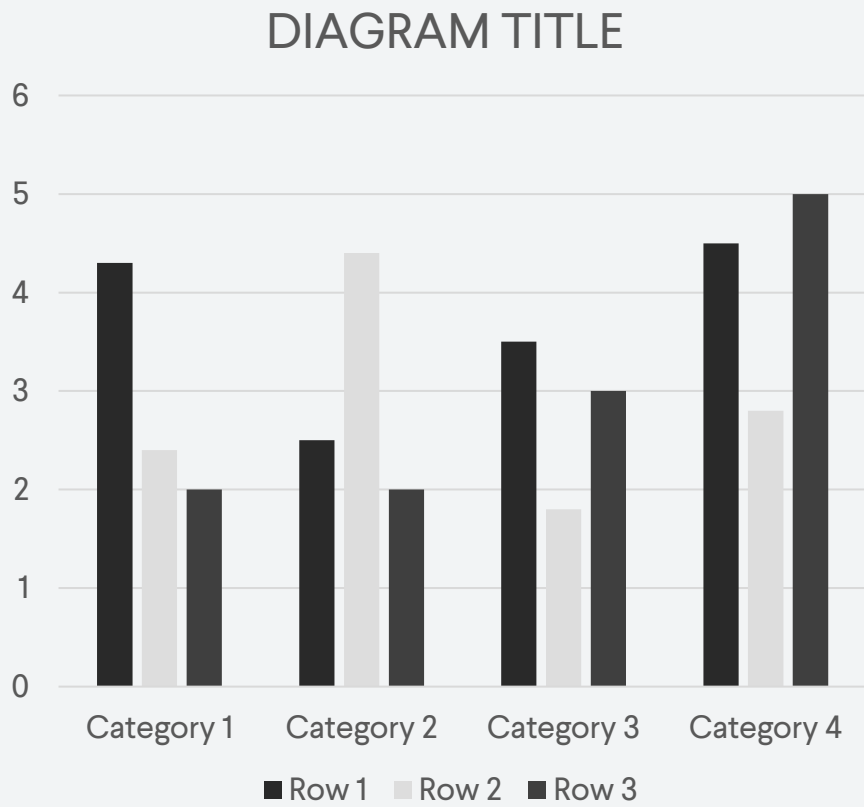
WHAT TYPES OF
VISUALIZATIONS DO YOU
KNOW?



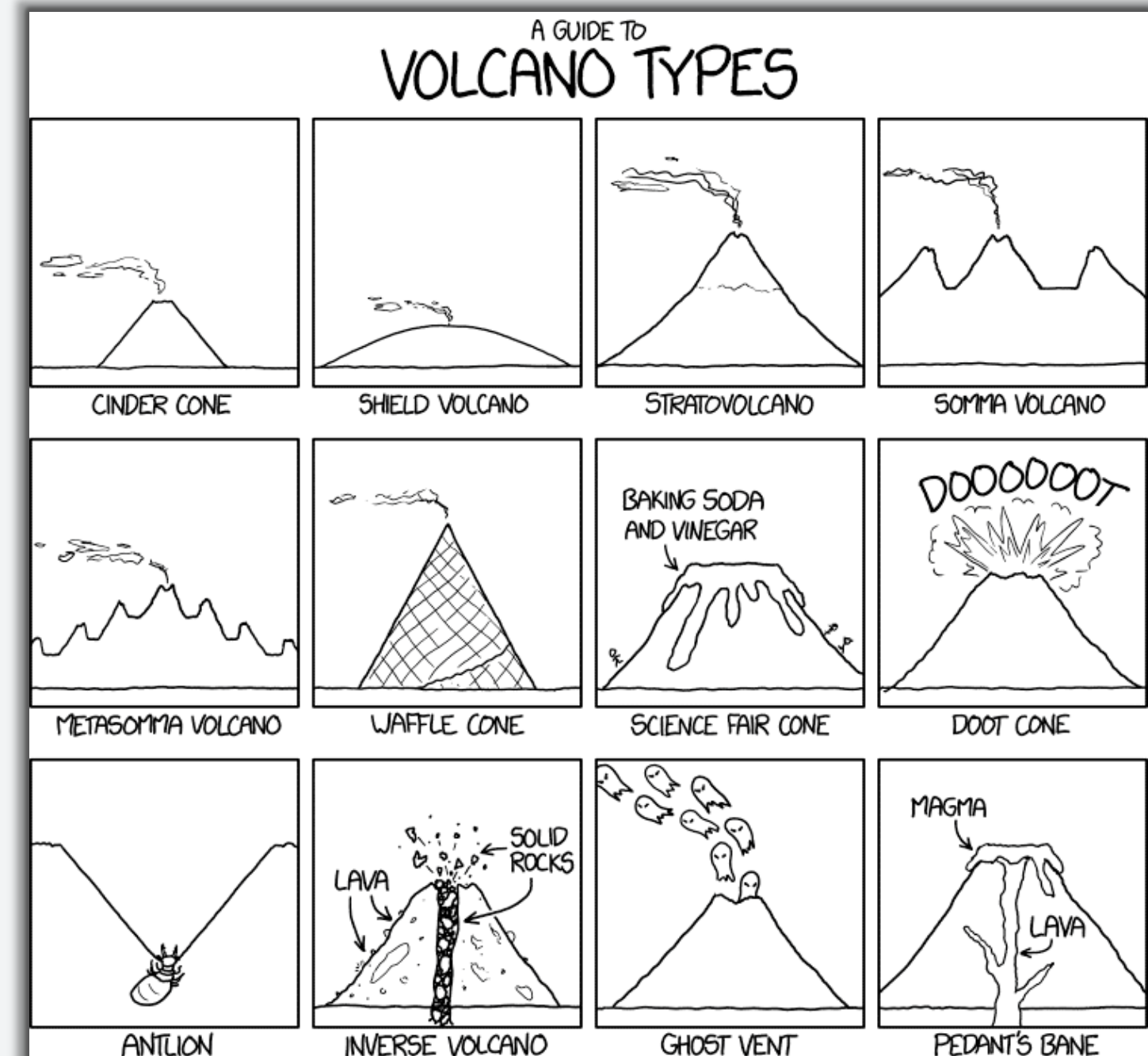
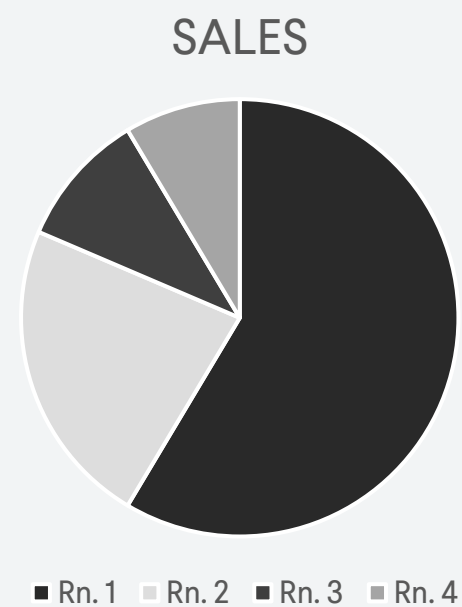
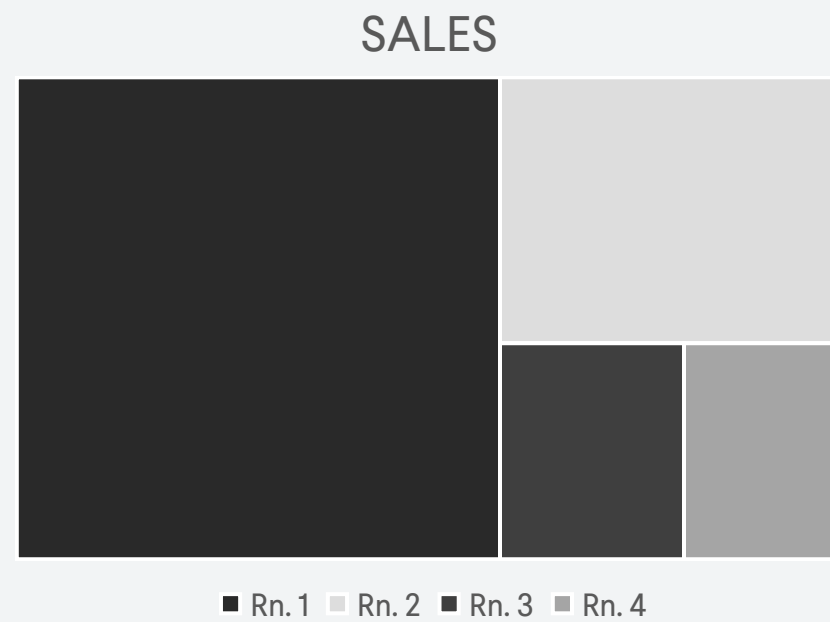
BASIC APPROACHES TO VISUALIZATION

DATA TYPE	SUITABLE TYPES OF VISUALIZATION
Univariate data	Histograms, box plots
Two-dimensional data	Scatter plots, heat maps, adjacency matrices
Multidimensional data	PCA, t-SNE, parallel coordinates
Time series	Line charts, seasonality charts
Geographic data	Maps, coordinate graphs

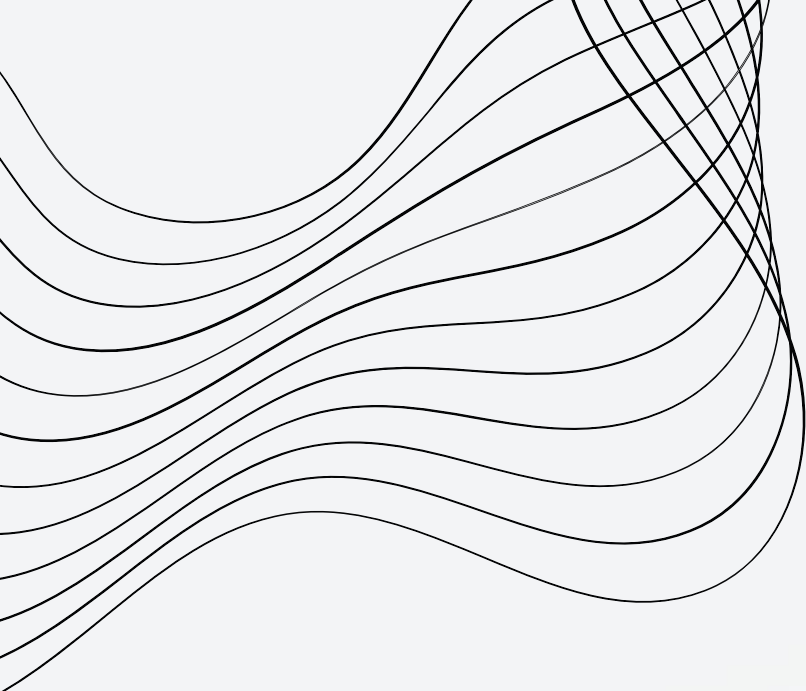
NUMERICAL DATA



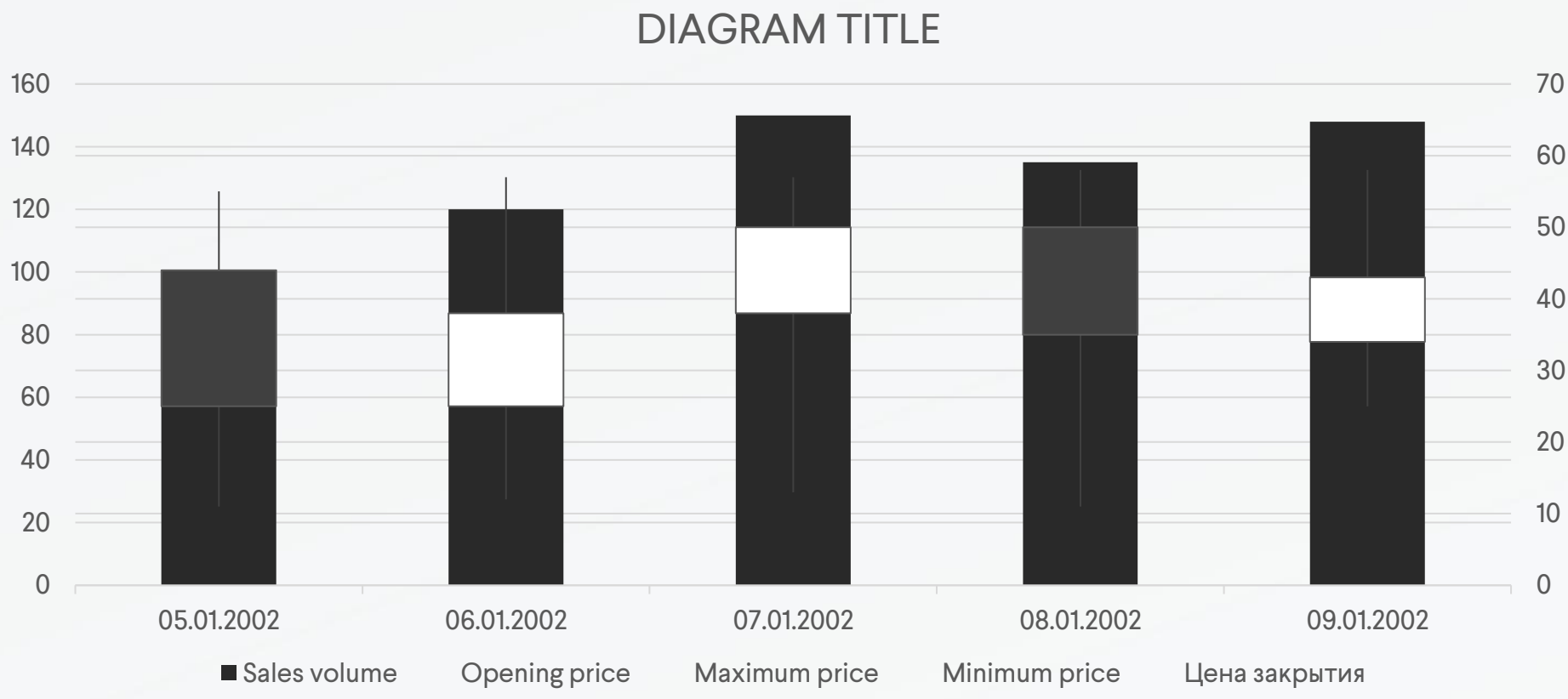
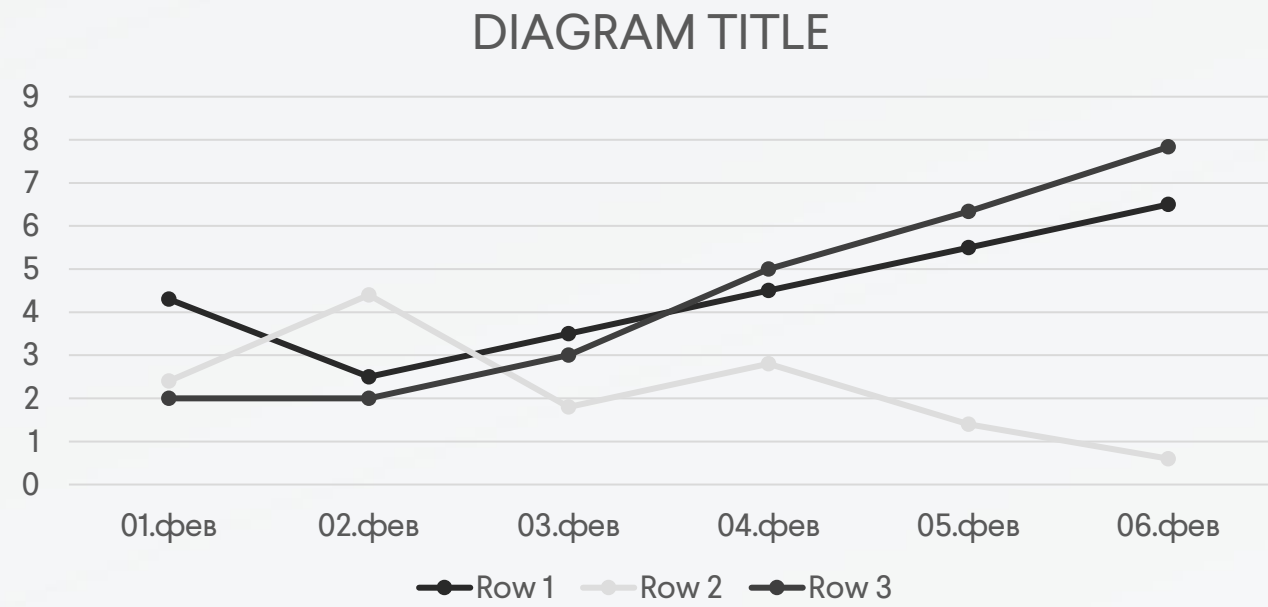
CATEGORIES



by Prof. Katharine Hayhoe, xkcd.com



T I M E S E R I E S

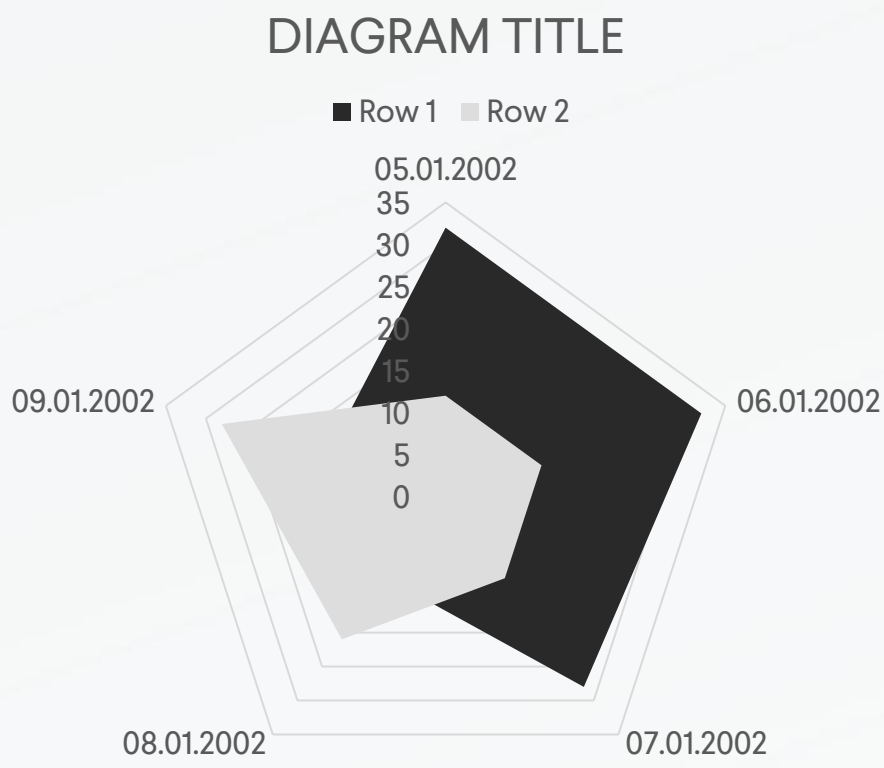
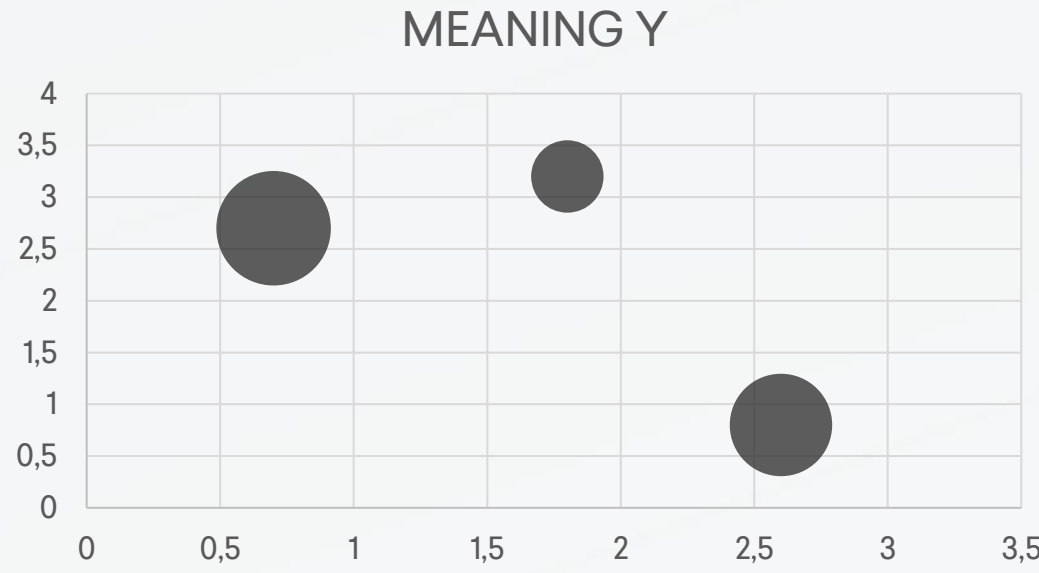
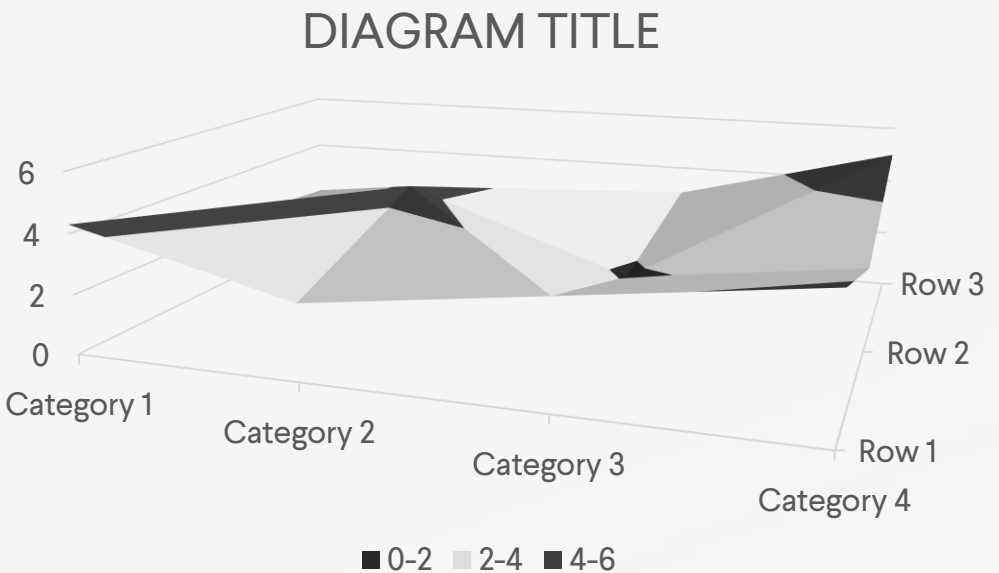


WHERE EACH COUNTRY SHOULD BE, BASED ON ITS TIME ZONE(S)



G E O G R A P H I C A L D A T A

MULTIDIMENSIONAL DATA



QUIZ

What data type are these
objects?



QUIZ “DATA TYPES”

**List of specialties at
the university**

Text/categorical data

Food energy table

Numerical
(multidimensional) data

IQ test results

Numerical (one-
dimensional) data

**Placements during
50 min race**

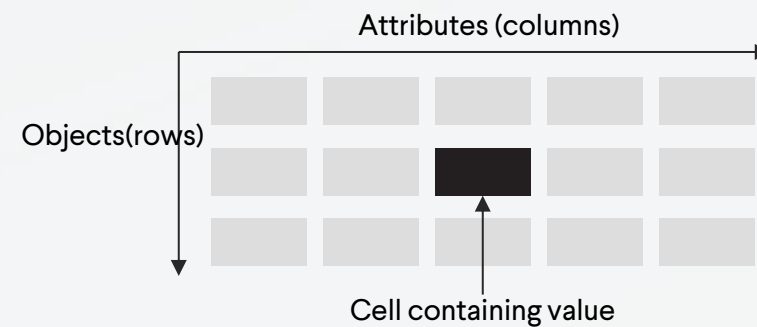
Time series

**Amazon product
ratings**

Numeric/categorical data

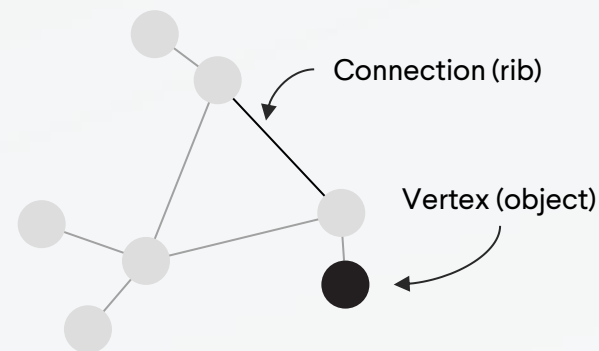
TYPES OF DATASETS

TABLES



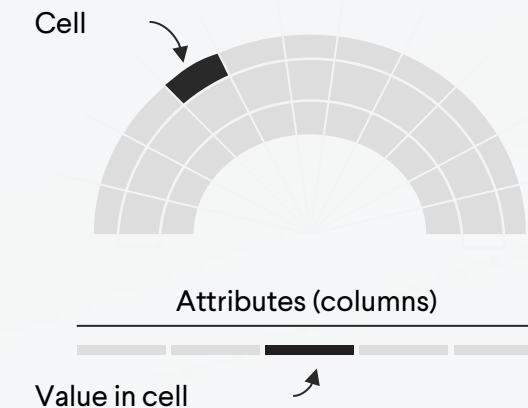
QuotaAmount,OwnerName,Username
150000,Chris Riley,tra.ub@example.com
150000,Chris Riley,tr.ub@example.com
150000,Chris Riley,trail.ub@example.com

GRAPHS



E1,E2,Weight
A1,A2,31.5
A1,Cb32,7.8
D3,B18,0.01

SEQUENTIAL DATA



QuotaAmount,150000,150000,1
50000,OwnerName,ChrisRiley,
ChrisRiley,ChrisRiley,Usernam
e,tra.ub@example.com,tr.ub@
example.com,trail.ub@exampl
e.com

GEOMETRY



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  },  
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LIST OF REFERENCES

